Estimation of the Warfarin Dose using Linear Multi-armed Bandits

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Abstract

In this project we aim to predict the optimal dosage for patients in an online setting. We assume that the expected reward is linear to the features (contexts) of the patients and formulate the problem as a 3-armed bandit problem. We implement and discuss two contextual multi-armed bandit algorithms, LinUCB with disjoint linear models and Thompson Sampling. It is shown that both UCB algorithms beat the two baselines consistently in terms of the cumulative regret, and outperform the naive baseline in terms of the total fraction of incorrect predictions.

1. Introduction

Warfarin is an anticoagulant that is used to treat or prevent blood clots in veins or arteries, which can reduce the risk of stroke, heart attack, or other serious conditions. The dosage of warfarin should be prescribed with caution, because warfarin increases the patients' risk of severe or fatal bleeding, especially if they have certain medical conditions, if 65 or older, or if they have had a stroke, or bleeding in the stomach or intestine. In addition, many other drugs can increase the risk of bleeding when used with warfarin. [1] Thus, the doctors must be very careful when deciding the dosage for the patients.

In this project, we aim to predict the best dosage for patients in an online setting, where we learn the optimal dosage upon observing the outcome of giving a certain dosage to each patient. Contextual multi-armed bandit problem comes as a natural setup for this problem because we can think of a certain dosage as an arm, and the information of a patient as the context. In Section 3 we will describe two baselines and implement two very popular contextual multi-armed bandit algorithms, LinUCB with disjoint linear models and Thompson Sampling. We discuss and compare all the algorithms in Section 4 and show that the two UCB algorithms beats both baselines consistently in terms of the cumulative regret, and outperforms the naive baseline in terms of the total fraction of incorrect predictions.

[1] https://www.drugs.com/warfarin.html

1.1. Dataset

We use a publicly available patient dataset that was collected by staff at the Pharmacogenetics and Pharmacogenomics Knowledge Base (PharmGKB) (Consortium, 2009) for 5700 patients who were treated with warfarin from 21 research groups spanning 9 countries and 4 continents. The dataset is in the tabular form where each row is an individual patient and the columns contains the information (features) about the patients such as demographics, background, phenotypes and genotypes. The dataset also contains the true patient-specific optimal warfarin doses (mg/week) for 5528 patients (we will ignore the 173 patients with missing doses), which are what the algorithm is going to predict. We will use 8 features as the Clinical Warfarin Dosing Algorithm does and we preprocess the features in the same way.

1.2. Problem Formulation

Let T be the number of time steps. At each time step t, a new patient arrives and we observe its individual feature vector $x_t \in R^d$: this represents the available knowledge about the patient (e.g., gender, age, ...). The decision-maker has access to K arms, where the arm represents the warfarin dosage to provide to the patient. For simplicity, we discretize the actions into $K = 3$.

- Low warfarin dose: under 21 mg/week
- Medium warfarin dose: 21-49 mg/week
- High warfarin dose: above 49 mg/week

If the algorithm identifies the correct dosage for the partient, the reward is 0, otherwise a reward of -1 is received. We assume that the reward for each arm depends on the patient features. For simplicity, consider a linear model where the reward for arm $i \in [K]$ for a patient with features X_t is

$$r_t(x_t, a_i) = x_t^T \beta_i + \epsilon_{i,t} \tag{1}$$

where $\beta_i \in R^d$ is an unknown parameter and the $\epsilon_{i,t}$ are independent Gaussian random variables with zero mean.

The goal is to come up with an algorithm A that learns a mapping $\pi : X \to a$ that yields the maximal expected reward. Let $\pi_t(s) \in [K]$ denote the arm chosen by policy π at

time $t \in [T]$ for the patient s_t. Define the optimal policy π^* to be the policy that maximizes the expected reward across patients given the true β_j parameters for $j = 1, ..., K$: this algorithm always chose the arm with $\max_j(x_t^T \beta_j)$. Under the linear model, if the agent chooses arm i at timestep t it will incur the expected regret of

$$E[\max_j[x_t^T \beta_j] - x_t^T \beta_i]$$

Our goal is to create and evaluate algorithms that minimize the cumulative expected regret $R_T = \sum_{t=1}^{T} E[_j[x_t^T \beta_j] - x_t^T \beta_i]$

2. Related Work

There have been a lot of literature around multi-armed bandit algorithms. The context-free K-armed bandit problem has been studied by statisticians for a long time (Berry & Fristedt, 1985). One of the simplest and most straightforward algorithms is ϵ-greedy, which has an positive chance of randomly selecting an arm in order to balance the exploitation and exploration. In contrast to the unguided exploration strategy adopted by ϵ-greedy, another class of algorithms generally known as upper confidence bound algorithms (Agrawal, 1995) use a smarter way to balance exploration and exploitation where they select the arm that achieves a highest upper confidence bound (UCB for short). With appropriately defined confidence intervals, it can be shown that such algorithms have a small total T-trial regret that is only logarithmic in the total number of trials T, which turns out to be optimal.

The two contextual LinUCB algorithms implemented in this project are from (Li et al., 2010) and (Agrawal & Goyal, 2013). (Li et al., 2010) models personalized recommendation of news articles as a contextual bandit problem, a principled approach in which a learning algorithm sequentially selects articles to serve users based on contextual information about the users and articles, while simultaneously adapting its article-selection strategy based on user-click feedback to maximize total user clicks.

(Agrawal & Goyal, 2013) design a contextual UCB algorithm that falls under the class of bandit algorithms that are based on Bayes rule. (Agrawal & Goyal, 2013) design and analyze a generalization of Thompson Sampling algorithm for the stochastic contextual multi-armed bandit problem with linear pay-off functions. With appropriately defined prior distributions, Bayesian approaches may have good performance. However, these methods require extensive offline engineering to obtain good prior models (Li et al., 2010).

3. Approach

3.1. Baseline 1: Fixed-dose Approach

We will use two baselines. The first one is a naive Fixed-dose approach where 35mg/day (medium) dosage is given to every patient.

3.2. Baseline 2: Warfarin Clinical Dosing Algorithm

The second baseline is the Warfarin Clinical Dosing Algorithm [2], which is a linear model based on age, height, weight, race and medications that the patient is taking. This model was designed by (Consortium, 2009) after the authors tested a wide range of statistical modeling and machine learning algorithms on the $80-20$ train-test split of the entire dataset. Mean absolute error (MAE) and R-squared were used for evaluation and the linear model was selected because of its optimal performance and its few number of variables. The linear coefficients for the 8 features are shown in Figure 1.

Warfarin clinical dosing algorithm		
	4.0376	
-	0.2546 x	Age in decades
+	0.0118 x	Height in cm
+	0.0134 x	Weight in kg
-	0.6752 x	Asian race
+	0.4060 x	Black or African American
+	0.0443 x	Missing or Mixed race
+	1.2799 x	Enzyme inducer status
-	0.5695 x	Amiodarone status
=	**Square root of weekly warfarin dose****	

Figure 1. Coefficients of the Warfarin Clinical Dosing Algorithm.

3.3. Linear UCB with Disjoint Linear Models

We implement the LinUCB with Disjoint Linear Models as described in (Li et al., 2010). Denote $X_{1:t,i}$ be the design matrix of dimension $m \times d$ at trial t, whose rows correspond to m training inputs (e.g. m patients that are observed previously for arm i). Let $c_i \in R^m$ be the corresponding response vector (e.g. the corresponding $-1/0$ rewards).

We assume the expected payoff of an arm $i \in [K]$ is linear in its feature X_t as in Section 1.2. This model is called disjoint because the parameters are not shared among different arms. That is to say, we have three linear models for each arm, the training data used for fitting each linear model is accumulating overtime as the patients arrive and specific arms are selected. After we select an arm for each patient and observes the reward, we apply ridge regression to $X_{1:t,i}$ and c_i to re-estimate the coefficients in the linear model.

$$\hat{\beta}_i = (X_{1:t,i}^T X_{1:t,i} + I_d)^{-1} X_{1:t,i}^T c_i \tag{2}$$

where I_d is the $d \times d$ identity matrix. When components

[2] S1f of appx.pdf

Algorithm 1 LinUCB with disjoint linear models

Input: $\alpha \in R_+$
for $i \in [K]$ **do**
  $A_i \leftarrow I_d$
  $c_t \leftarrow 0_{d\times 1}$
end for
for $t = 1, 2, 3, ..., T$ **do**
  Observe features of patient who arrives: $x_t \in R^d$
  for $i \in [K]$ **do**
    $\hat{\beta}_i \leftarrow A_i^{-1} c_i$
    $p_i = x_t^T \hat{\beta}_i + \alpha\sqrt{x_t^T A_i^{-1} x_t}$
  end for
  Choose arm $a_t = argmax_{i\in[K]} p_i$ with ties broken arbitrarily. Observe a reward $r_t \in \{-1, 0\}$.
  $A_i \leftarrow A + x_t x_t^T$
  $c_i \leftarrow c_i + r_t x_t$
end for

in c_i are independent conditioned on corresponding rows in $X_{1:t,i}$, it can be shown (Walsh et al., 2012) that, with probability at least $1-\delta$,

$$|x_t^T \hat{\beta}_i - E[r_{t,a}|x_t]| \leq \alpha\sqrt{x_t^T (X_{1:t,i}^T X_{1:t,i} + I_d)^{-1} x_t} \quad (3)$$

for any $\delta > 0$ and $x_t \in R^d$, where $\alpha = 1 + \sqrt{ln(2/\delta)/2}$ is a constant. In other words, the inequality above gives a reasonably tight UCB for the expected payoff of arm a, from which a UCB-type arm-selection strategy can be derived: at each trial t, choose

$$a_t = argmax_{i\in[K]}(x_t^T \hat{\beta}_i + \alpha\sqrt{x_t^T A_i^{-1} x_t}) \quad (4)$$

where $A_i = X_{1:t,i}^T X_{1:t,i} + I_d$. Note that our notations are slightly different from (Li et al., 2010). The main difference is that we assume x_t is the same across all candidate arms, whereas in (Li et al., 2010) they use $x_{t,i}$ which allows different features for $i \in [K]$. The algorithm is shown in 1.

3.4. Linear UCB with Thompson Sampling

We also consider another popular algorithm proposed by (Agrawal & Goyal, 2013). Thompson sampling is a randomized algorithm based on Bayesian ideas, and has generated significant interest. Since we assume that the reward is a linear function of the contexts with Gaussian noise as in Eq. 1, we can assume that the likelihood of reward $r_i(t)$ can be given by the pdf of Gaussian distribution $\mathcal{N}(x_t^T \mu, v^2)$, where $v = R\sqrt{\frac{24}{\epsilon} d \ln \frac{1}{\delta}}$ (the parameters will be defined later). Using Baye's rule, we know that if the prior of for μ at time t is given by $\mathcal{N}(\hat{\mu}_t, v^2 A^{-1})$, then the posterior distribution upon observin $r_i(t)$ is

$$Pr(\hat{\mu}|r_i(t)) \propto Pr(r_i(t)|\hat{\mu}) Pr(\hat{\mu}) \quad (5)$$

Algorithm 2 LinUCB with Thompson Sampling

Input: $R, \epsilon, \delta \in R_+$
Set $v = R\sqrt{\frac{24}{\epsilon} d \ln \frac{1}{\delta}}$.
for $i \in [K]$ **do**
  Set $A_i = I_d$, $\hat{\mu}_i = 0_d$, $f_i = 0_d$.
end for
for $t = 1, 2, 3, ..., T$ **do**
  Observe features of patient who arrives: $x_t \in R^d$
  Sample $\hat{\mu}_i(t)$ from distribution $\mathcal{N}(\hat{\mu}_i, v^2 A_i^{-1})$.
  Choose arm $a_t = argmax_{i\in[K]} x_t^T \hat{\mu}_i(t)$ with ties broken arbitrarily. Observe a reward $r_t \in \{-1, 0\}$.
  $A_i \leftarrow A_i + x_t x_t^T$
  $f_i \leftarrow f_i + x_t r_t$
  $\hat{\mu}_i \leftarrow A_i^{-1} f$
end for

At each timestep, we simply select the arm that maximizes $x_{i,t}^T \hat{\mu}(t)$, as illustrated in Algorithm 2.

Regret Bound

In addition, with some additional assumptions, we can bound the regret asymptotically.

- The gaussian noises $\epsilon_{i,t}$ in Eq. 1 are conditionally R-sub-Gaussian for a constant $R \geq 0$, which is satisfied if $\forall t, r_t \in [x_t^T \mu - R, x_t^T \mu + R]$.
- Assume that $||x_t|| < 1$, $||\mu|| < 1$, $||\Delta(t)|| < 1$ in order to make the regret bounds scale-free. Δ is the regret. So we will normalize the contexts x_t.

Under the above assumptions, it is shown in (Agrawal & Goyal, 2013) that with probability $1-\delta$, the total regret in time T is bounded by $O(d\sqrt{\frac{NT^{1+\epsilon}\ln N}{\epsilon}(\ln T \ln \frac{1}{\delta})})$ for any $0 < \epsilon < 1$, $0 < \delta < 1$, where N is the number of different parameters.

4. Experiments Results

4.1. Evaluation Metrics

For each algorithm, we report two measures of performance. *Regret* as defined in Section 1.2 and *the fraction of incorrect dosing decisions*, the fraction of incorrect dosing decisions made so far by the algorithm at the current timestep out of all patients seen so far by the algorithm at the current timestep.

4.2. Results

For Algorithm 1and Algorithm 2, we use $\alpha = 0.5$ and $v = 0.2$ correspondingly. To guarantee robustness, we run all the algorithms on 20 random permutations of patients

and plot the performance with 95% confidence intervals (T-distribution).

On the entire dataset (5528 patients), the Fixed-dose algorithm has a 39.7% error rate (the fraction of incorrect dosing decisions), and the Warfarin Clinical Dosing Algorithm has a 36.5% error rate. The total fraction of incorrect doses for all algorithms are shown in Table 1. Note that the numbers for the baselines are deterministic, whereas they can vary for the LinUCB algorithms depending on the order of the patients arriving. The 95% confidence intervals for the LinUCB algorithms are shown with $\pm$ signs. In addition, we plot the running fraction of incorrect decisions in Figure 2. We see that the Clinical Dose algorithm is consistently better than all the other three models, which is because this model is fitted on the entire dataset. That is to say, it is cheating because it looks at the entire dataset whereas our LinUCB algorithms are making decisions after we see each patient. It makes sense for the LinUCB models to do much worse in the beginning, but overall, they have approximately the same total error rate as the Clinical Dose algorithm.

The cumulative regret for all algorithms are shown in Figure 3. Note that both baselines have linear regret whereas the LinUCB algorithms have sub-linear regret, which are expected by the regret bound analysis in (Li et al., 2010) and (Agrawal & Goyal, 2013). We can see that both LinUCB models have very low total regret, although the confidence intervals around the regret curve can be wide. This is because the amount of "exploration", an important factor that makes our UCB algorithms achieve great performance, can have very difference effects across different randomization of the dataset.

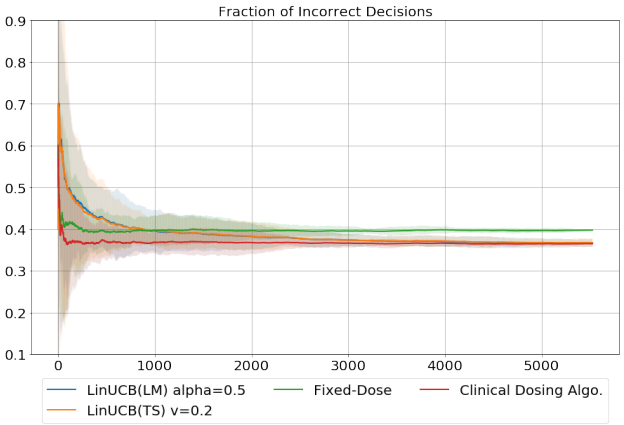

Figure 2. The fraction of incorrect dosing decisions made so far by the algorithms at the current timestep out of all patients so far by the algorithm at the current timestep. The 95% confidence interval (t-distribution) is also plotted.

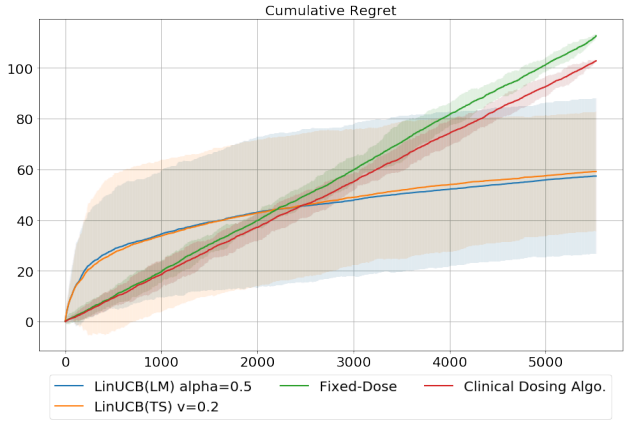

Figure 3. Cumulative regret for all models with confidence bands. Both baselines have linear regret whereas the LinUCB algorithms have sub-linear regret.

Table 1. Total fraction of incorrect doses for different algorithms. Note that the Fixed-dose and Clinical Dose algorithms give deterministic numbers on the entire dataset.

	R_T	INCORRECT FRAC.
FIXED-DOSE	112.59	39.7
CLINICAL DOSE	102.73	36.5
LINUCB	57.32 ± 10.98	36.6 ± 0.01
LINUCB(TS)	59.13 ± 8.46	36.7 ± 0.01

4.3. Sensitivity Analysis

In our experiments, we noticed that the performance of our LinUCB algorithms are very sensitive to the coefficient of the upper confidence bound (α for Algorithm 1 and v for Algorithm 2). We plot the cumulative regret for both algorithms using different coefficients in Figure 4 and Figure 5.

In the original papers (Li et al., 2010) (Agrawal & Goyal, 2013), the authors gave the theoretical choices of the coefficients such that the sublinear regret bound is achieved.

- For Algorithm 1, $\alpha = 1 + \sqrt{\ln(2/\delta)/2} = 2.23$ with $\delta = 0.1$.
- For Algorithm 2, $v = R\sqrt{\frac{24}{\epsilon} d \ln \frac{1}{\delta}} = 2.15$ with $R = 0.1$, $\epsilon = 0.95$, $d = 8$ and $\delta = 0.1$.

Nonetheless, we we show in Figure 4 and Figure 5, the above choices give much worse total regret compared to the choices we are using $\alpha = 0.5, v = 0.2$, although they all have sublinear regret.

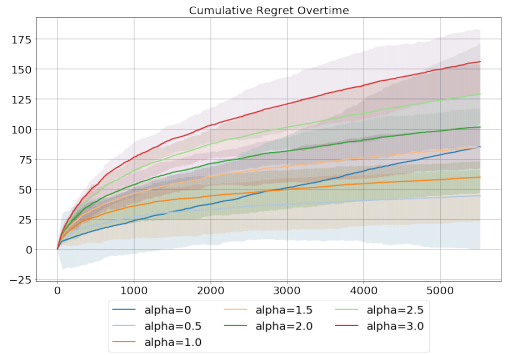

Figure 4. Cumulative regret for LinUCB vs. α.

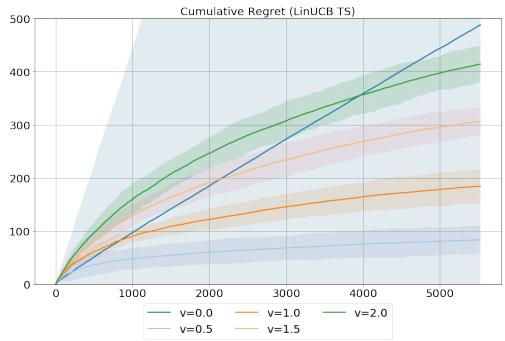

Figure 5. Cumulative regret for LinUCB (TS) vs. v.

4.4. Linear Models

In this section, we take a deeper look at the linear models in the two LinUCB models.

The coefficients of the 3 linear models $\hat{\beta}_i \in R^8$ in LinUCB with disjoint linear models are shown in Figure 6. The $\hat{\mu}_i \in R^8$ in LinUCB with TS are in Figure 7. An intuitive way to interpret these coefficients is to say that for a patient with positive features for the positive signed coefficients for arm i are likely to be given dose i. For instance, from Figure 6 we can tell that a senior Asian patient who takes Amiodarone but no Enzyme inducer is likely to be given a low dosage.

Another observation that can serve as a sanity check of the algorithm is that antithetical features have opposite signs. For instance, the signs for IsAsian and IsAfrican features are clearly opposite to each other for all three arms.

One assumption we made in Section 1.2 is that the reward for each arm can be modelled as a linear model of the patient features with Gaussian noises. But is that the case in our dataset? We plot the prediction errors for each arm in our LinUCB models in Figure 8. We see that the errors seem

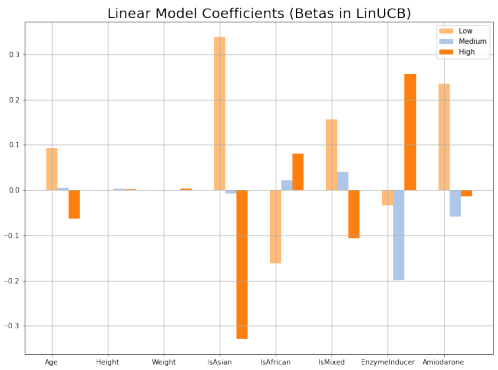

Figure 6. Coefficients of the 3 linear models in LinUCB with disjoint linear models.

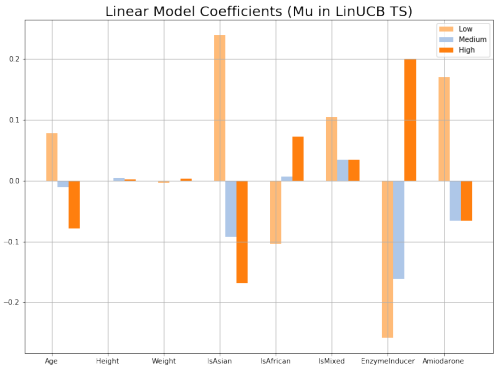

Figure 7. Mean estimates in LinUCB with Thompson Sampling.

to be approximately normally distributed around two peaks since the reward observed are either 0 or -1.

5. Conclusion

In this project, we implemented and showed that the two UCB algorithms beat both baselines consistently in terms of the cumulative regret, and outperforms the naive baseline in terms of the total fraction of incorrect predictions. Since the Clinical Dose model was fitted on the entire dataset (hence not online), it would makes sense for a online algorithm like LinUCB to not perform as well. However, we showed that the LinUCB algorithms in an online setting still achieve decent performance.

In addition, we observed that the performance of LinUCB is very sensitive to the multiplier of the confidence bound. Thus, it is crucial to choose a good multiplier since the penalty of giving a wrong dose to patients can be severe in real world. However, in reality, we won't know what an optimal multiplier choice before we see the actual data, we

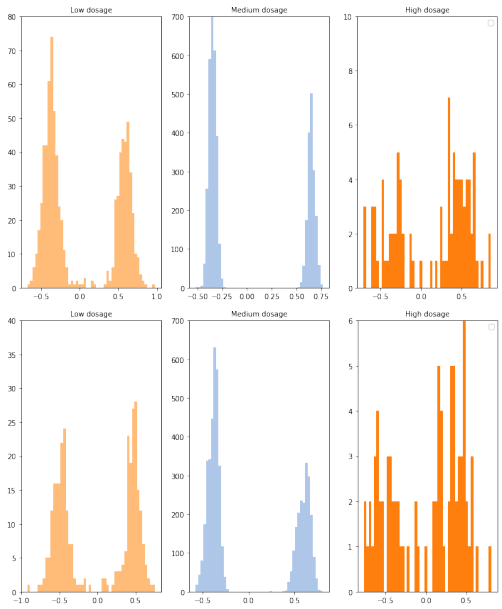

Figure 8. Errors of the predictions for each arm. Top: LinUCB with linear models. Bottom: LinUCB with TS.

have to guess and adjust as we go.

For future work, we could explore algorithms that are more robust to the choice of confidence bound multipliers. Since the multipliers are indicative of how much we want to explore vs. exploit, it would also be interesting to see what the deciding factors are that makes a dataset sensitive to the multipliers.

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